PROJECT AND TEAM INFORMATION

Project Title

DNA (Destructured Neural Agent)

Student / Team Information

Team Name: Evowalk *Team #DSCPP-III-2026-T051*	*Avengers*
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Mahar, Tanishka – 2510011047* *mahartanishka068@gmail.com* 
Team member 2 *(Last Name, name: student ID: email, picture):*	*Khati, Mansha – 2510380049* *manshaakhatiii@gmail.com* 
Team member 3 *(Last Name, name: student ID: email, picture):*	*Juglan, Alankrit – 2510350082* *alankritjuglan@gmail.com* 
Team member 4 *(Last Name, name: student ID: email, picture):*	*Vinamra – 2510012765* *victorvc3018@gmail.com* 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

One fine day, I woke up thinking about life that day. I realized that everything goes back to the strands, what we call DNA.

DNA can be anything and everything which occupies this world. So when I got this fine opportunity of making a project based on anything that I could have, I chose the one exact thing that occupied my mind all the time which evidently was DNA.
Now what DNA is to me Can it be the same to anyone else?
*The meaning that I gave this thought was *Destructured Neural Network* now how my larp is going to work is:-*

Robots can face unexpected obstacles. The final project will be a C++ graphical simulation. It will contain a virtual robot and obstacles.A fixed movement strategy may not work in every environment and a small change in the environment can affect the robot's path. Our project explores this problem using a software simulation. This contains a virtual robot and obstacles.

Our OWN Genetic Algorithm we call it The SAUCE if you catch our drift you know what we mean, will be used to improve the robot's movement strategies over multiple generations.

State of the Art / Current solution (1 pt)

While thinking about the genetic systems we stumbled across the very human nature of things. how gene operates, how the dna stands operate within each other they don't use something which is highly complex yet has been helping evolution for a long while.
We try to utilise what the nature have been using for a long while now, and we don't have to go through multiple hard equations nor did we have to use multiple complex solutions for a problem such as this.

With the power of our ancestors and the efforts of science, we had on our hands the Destructed neural network.

Robot navigation can use predefined rules or it can also use path-planning algorithms. But that's boring. These methods work well in known environments.... and also like changing the environment can make them less efective.

Our GA provides another approach. It starts with multiple possible solutions in the hood and then, these solutions are tested using a fitness function. Better solutions are selected. Crossover and mutation are then used to create new solutions. We will use this process for virtual robot navigation.

Project Goals and Milestones (2 pts)

Our main goal is to create a graphical C++ simulation (or others if possible) which will contain a virtual robot. The robot will try to reach a target. It will also need to avoid obstacles. Or we can reduce a limb and make it reiterate.
What is the true meaning of evolution that's our First point. Do we really understand the preciseness of our evolution tree? With this project our goal is to understand the complexities behind general neural algorithm and how they operate. Mixing the concepts of computer science, core and AI/ML to create a new abomination.

Milestones:
1. *Create the the Genetic Algorithm Prototype.*
2. *Implement the fitness function.*
3. *Implement selection, crossover, and mutation*
4. *Connect the movement strategies to an engine's robot*
5. *Add the graphical interface using SFML.*
6. *Test the robot across multiple generations.*

Project Approach (3 pts)

The project will be developed in C++. We will use Object-Oriented Programming. Data Structures will also be used.The Genetic Algorithm will generate movement strategies. Each strategy will be like one possible way for the robot to move. Then the robot will test them in the environment.

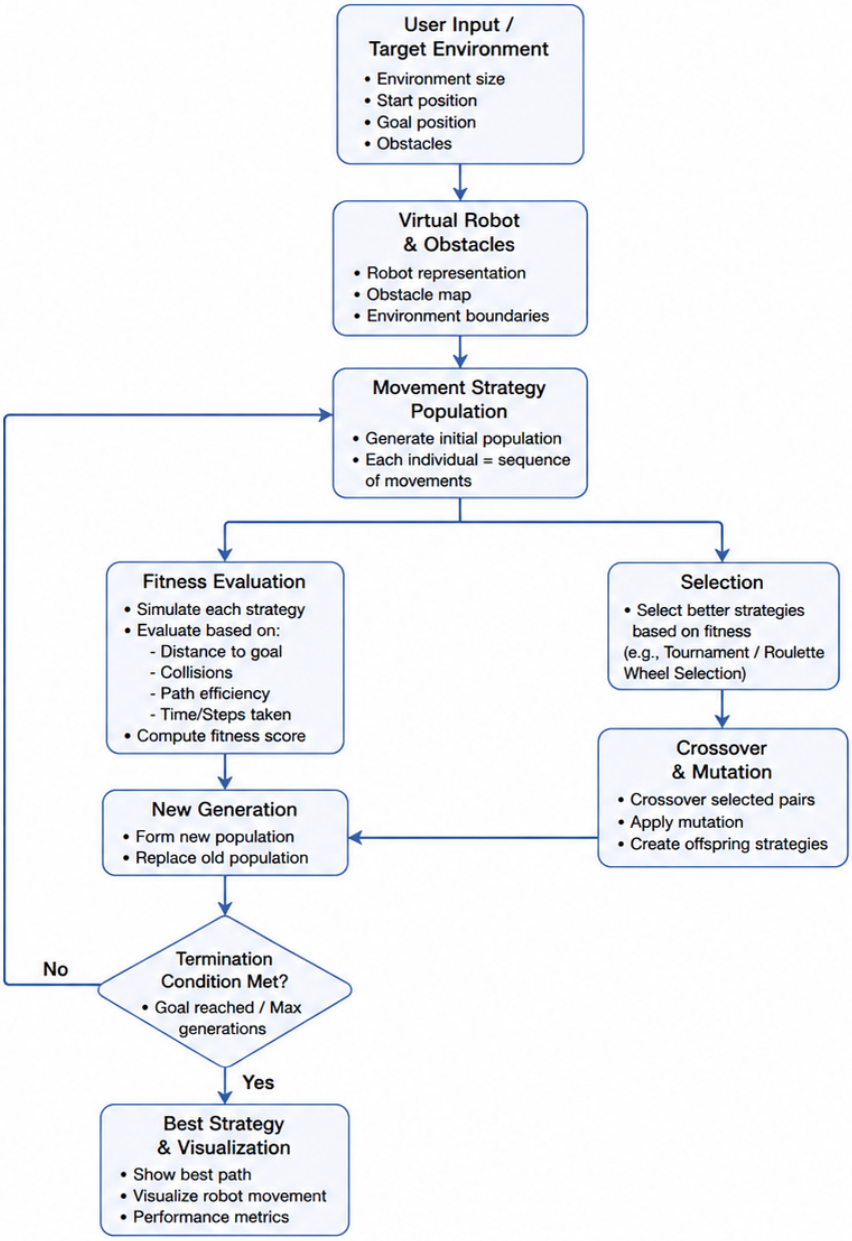
Each possibility will receive a fitness value. The value will show how well the robot performed. The best will be bred, and then Crossover will combine selected strategies.

While you see 'Mutation' will introduce small changes. The new children of the previous generations will form the next generation. This process will repeat for several generations. We will then compare the robot's performance.

SFML will be used for the graphical simulation. It will display the robot and obstacles. It will also show the robot's movement.
In our final project, when we insert our algorithm into the robot's brain, the robot will be allowed to make a movement and record it for that entire time frame.
For this instance, since our project is virtual, we can virtually set any time frame. If it had been real-life testing, we had to lose real-time seconds or even minutes.

But since it is virtual, we can adjust the time using the computer time. After the recording of the time frame, our algorithm will calculate the best possible outcome from the experiment and nudge its parameters a little bit further into the right direction.

System Architecture (High Level Diagram)(2 pts)



DNA - De-structured Neural Agent

Project Outcome / Deliverables (1 pts)

The data structure part was designed in C, later implemented in C++ and in the further models of this project we wish to use python which will give us the opportunity to expand its horizons beyond the limits of C++ in graphical areas.

For example, going into more game development Vibes or even making projects which are suitable with opencv library of python and working with cameras and iot devices in real time as compared to C++ which is quite restrictive when it comes to iot devices because it has a very restricted our types of libraries and it does not give us as much allowance to work with the models.

Although it is very fast but it doesn't allow us to experiment as much so far. Recent earlier models we will decide to stick to C++ so that our model can be as efficient as possible. But later when we will have time and we will, we won't have any hurry. We will be using different libraries and different languages so it can work under any architecture. Thank you for that. The final project will be a C++ graphical simulation. It will contain a virtual robot and obstacles. The robot will try to reach a target. The simulation will show different movement strategies. It will also show their progress across generations. The project will demonstrate Genetic Algorithms, Data Structures, and Object-Oriented Programming. Our project our algorithm works over multiple different architectures, including PCS with Linux, distributions, windows, architecture, or even Mac.
Also including Unix base systems like Android and what not because our project works on very lightweight libraries in-built libraries of C and C++ the data side.

Assumptions

1. *The engine will have fixed boundaries.*
2. *And it will contain predefined obstacles.*
3. *The robot will be simulated in software.*
4. *It will have a fixed target.*
5. *A fitness function will compare movement strategies.*
6. *No physical robot will be used hence no damage before Irl testing.*

References

1. *Genetic Algorithms and evolutionary computation literature.*
2. *C++ Standard Library documentation.*
3. *SFML Documentation.*
4. *Mutation Breeding*